Schematic Sheets

#	Name	File	Description
1	Main	SensWear_Main.SchDoc	Top-level schematic showing system hierarchy and interconnection between components
2	MCU	SensWear_MCU.SchDoc	Micro-Controller Unit (MCU)
3	Memory	SensWear_Memory.SchDoc	External EEPROM / non-volatile memory
4	Power	SensWear_Power.SchDoc	Battery input, voltage regulation, charging circuitry, and daughter board regulator
5	MPU	SensWear_MPU.SchDoc	Motion sensing section (IMU)
6	LED	SensWear_RGB.SchDoc	RGB/status LED circuitry
7	Debug	SensWear_Debug.SchDoc	Debug and development interface including UART and SWD/programming access
8	DaughterBoard	SensWear_DaughterBoard.SchDoc	Modular daughter-board interface exposing power, communication buses, and GPIOs

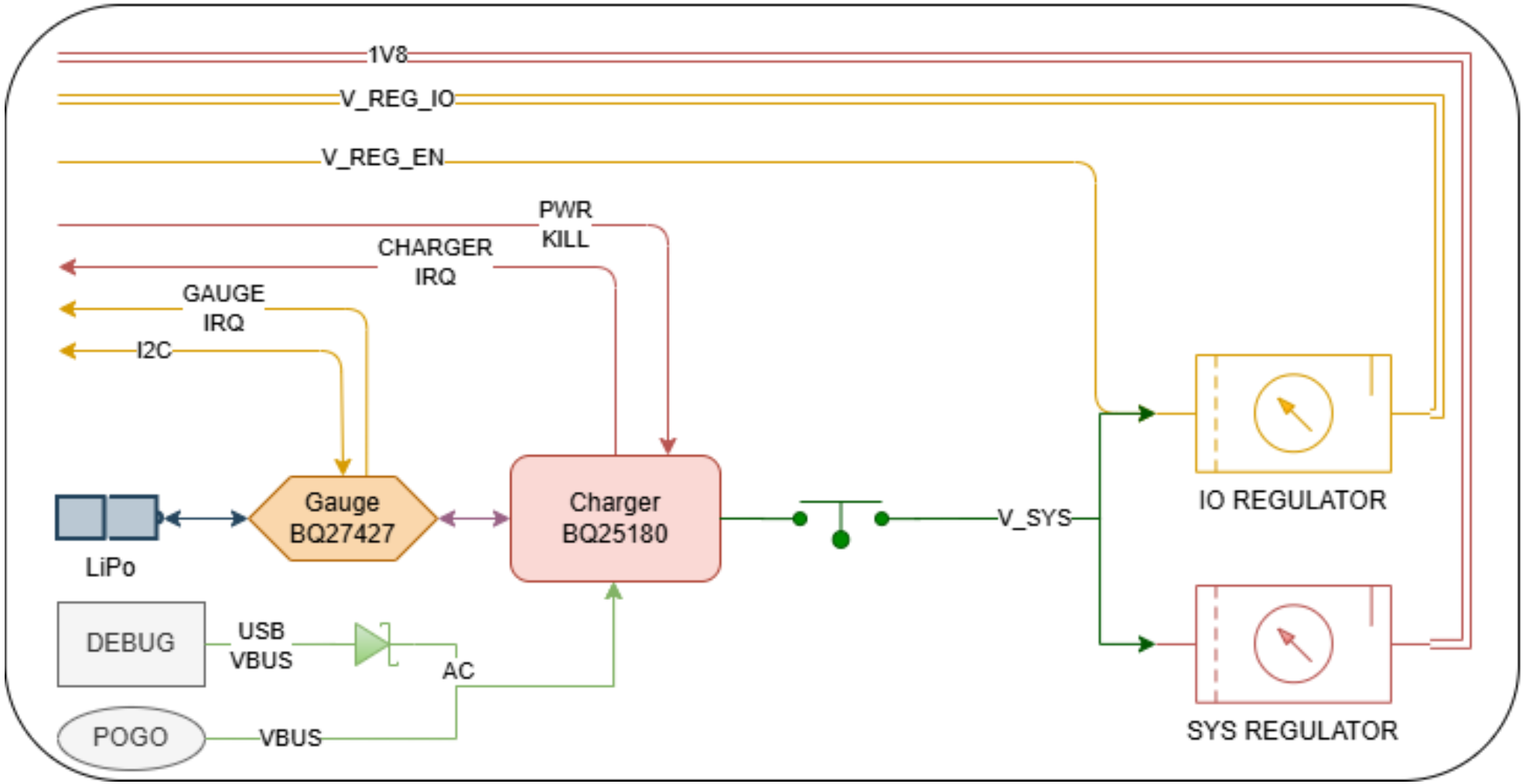
Revision History

Revision	Date	Description
A0	May 2025	Initial validation design on rigid PCB
B0	March 2026	Flexible PCB manufacturing updates
C0	August 2026	Battery placement related optimization. Turn On/Off Slide switch added.

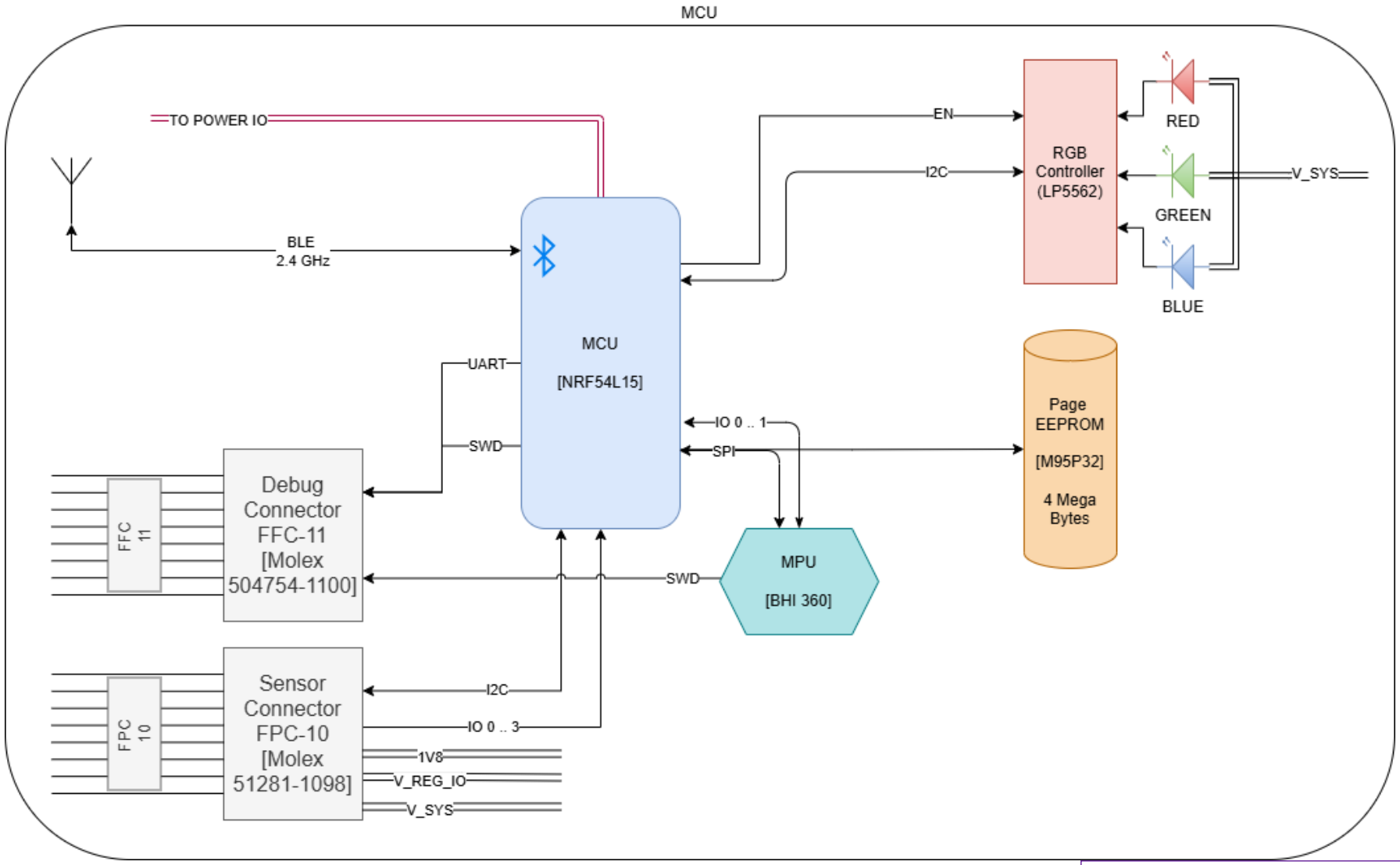
SensWear-V1R1

Designer/Engineer: Yusein Ali	Board Title: SensWear-Hub		
Drawn By: Yusein Ali	Document Name: SensWear-V1R1-Hub-		
Approved By: Dariush Salami	Ver.: 1	Rev.: C0	Date: 31.08.2026

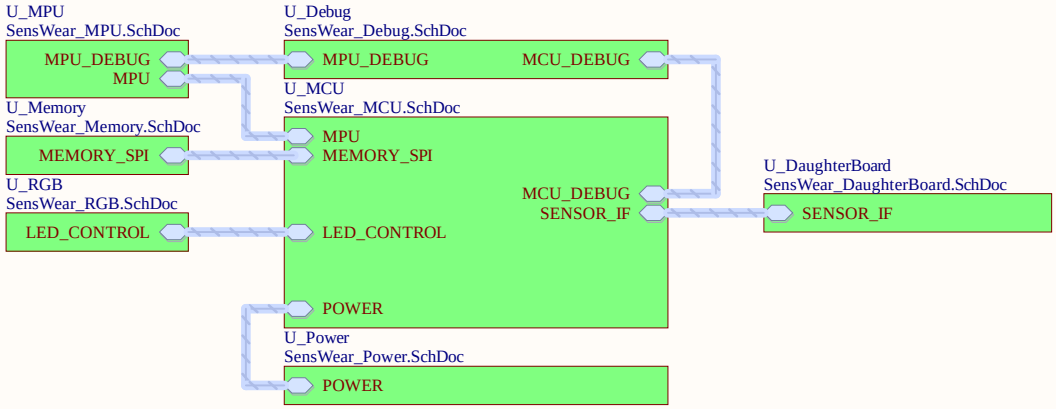
POWER



SensWear-V1R1			
Designer/Engineer Yusein Ali	**Board Title** SensWear-Hub		
Drawn By Yusein Ali	**Document Name** SensWear-V1R1-Hub-		
Approved By Dariush Salami	**Ver.** 1	**Rev.** C0	**Date** 31.08.2026



SensWear-V1R1			
Designer/Engineer Yusein Ali	**Board Title** SensWear-Hub		
Drawn By Yusein Ali	**Document Name** SensWear-V1R1-Hub-		
Approved By Dariush Salami	**Ver.** 1	**Rev.** C0	**Date** 31.08.2026



	SensWear-V1R1			
Designer/Engineer: Yusein Ali	Board Title: SensWear-Hub			
Drawn By: Yusein Ali	Document Name: SensWear-Main.SchDoc			
Approved By: Dariush Salami	Rev.: C0	Sheet: 1 of 8	Size: A4	
		Date: 31.08.2026		

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nRF54L15 PERIPHERAL PIN ASSIGNMENTS

[UART] - Instance: UART30 (Peripheral Domain)
RXD : P0.03 (Input)
TXD : P0.04 (Output)
NOTE: P0.03 and P0.04 are valid P0 digital I/O pins and should be used with UARTE30.

[SPI MASTER] - Instance: SPIM00 (MCU Domain - High Speed)
SCK : P2.01
MOSI : P2.02 (Dedicated IO0 Pad)
MISO : P2.04 (Dedicated IO1 Pad)
NOTE : These pins utilize dedicated hardware muxing.
Do not swap MOSI/MISO on these pins.

[I2C MASTER] - Instance: TWIM20 (Peripheral Domain)
SDA : P1.02
SCL : P1.03
REQ : External Pull-up Resistors required.

I2C ADDRESS TABLE (I2C21)
SDA: P1.02 | SCL: P1.03

Device Type	Part Number	Hex (7-bit)	Binary (8-bit with R/W=x)
Buck Converter	TPSM831021	0x68	0100 001x
Touch Controller	MTCH6102	0x25	0100 101x
LED Driver	LP5562	0x30	0110 000x
Temp Sensor	MAX30208	0x50	1010 000x
Fuel Gauge	BQ27427	0x55	1010 101x
Pulse Oximeter	MAX30101	0x57	1010 111x
Haptic Driver	DRV2605L	0x5A	1011 010x
Battery Charger	BQ25180	0x6A	1101 010x

NOTE: 'x' = 0 for WRITE, 1 for READ.
Ensure 1.5k pull-ups on SDA/SCL lines.

[CRYSTAL X2 NOTE]
Type: 32.768 kHz (CL=12.5pF)
Caps: INTERNAL (Firmware Enabled).
Do NOT place external capacitors.

```
&lfxo {
    status = "okay";
    load-capacitors = "internal";
    /* * Calculation for 12.5pF Crystal:
     * C_pin = 2 * (C_load - C_stray)
     * C_pin = 2 * (12.5pF - 4pF) = 17pF
     */
    load-capacitance-femtofarad = <17000>;
};
```

SENSOR INTERFACE (GENERIC)
Logic Level: 1.8V (V_REG_IO Domain)

[SHARED CONTROL BUS]
Interface : I2C (I2C21)
SDA : P1.02 (Pin 3)
SCL : P1.03 (Pin 4)
Pull-ups : 1.5kΩ to 1V8 (R15, R16) installed on board.
Note : P1.02 and P1.03 are NFC pins by default.
Disable NFC since these pins are used as I2C GPIO.

[SENSOR GPIOs / ANALOG INPUTS]
GPIO_0 : P1.14 / AIN7 (Pin 42)
- Capable of Analog Input (SAADC).
GPIO_1 : P1.13 / AIN6 (Pin 41)
- Capable of Analog Input (SAADC).
GPIO_2 : P1.12 / AIN5 (Pin 40)
- Capable of Analog Input (SAADC).
GPIO_3 : P2.11 / AIN4 (Pin 39)
- Capable of analog input (SAADC).

[POWER DOMAIN]
Supply : V_REG_IO (Switched Rail)
Control : V_REG_EN (P1.06) must be HIGH.
Note : Ensure sensors connected to these GPIOs are powered by V_REG_IO to avoid leakage currents when the sensor subsystem is OFF.

MCU (nRF54L15)DEBUG & DIAGNOSTIC INTERFACE
Logic Level: 1.8V (VDD_IO)

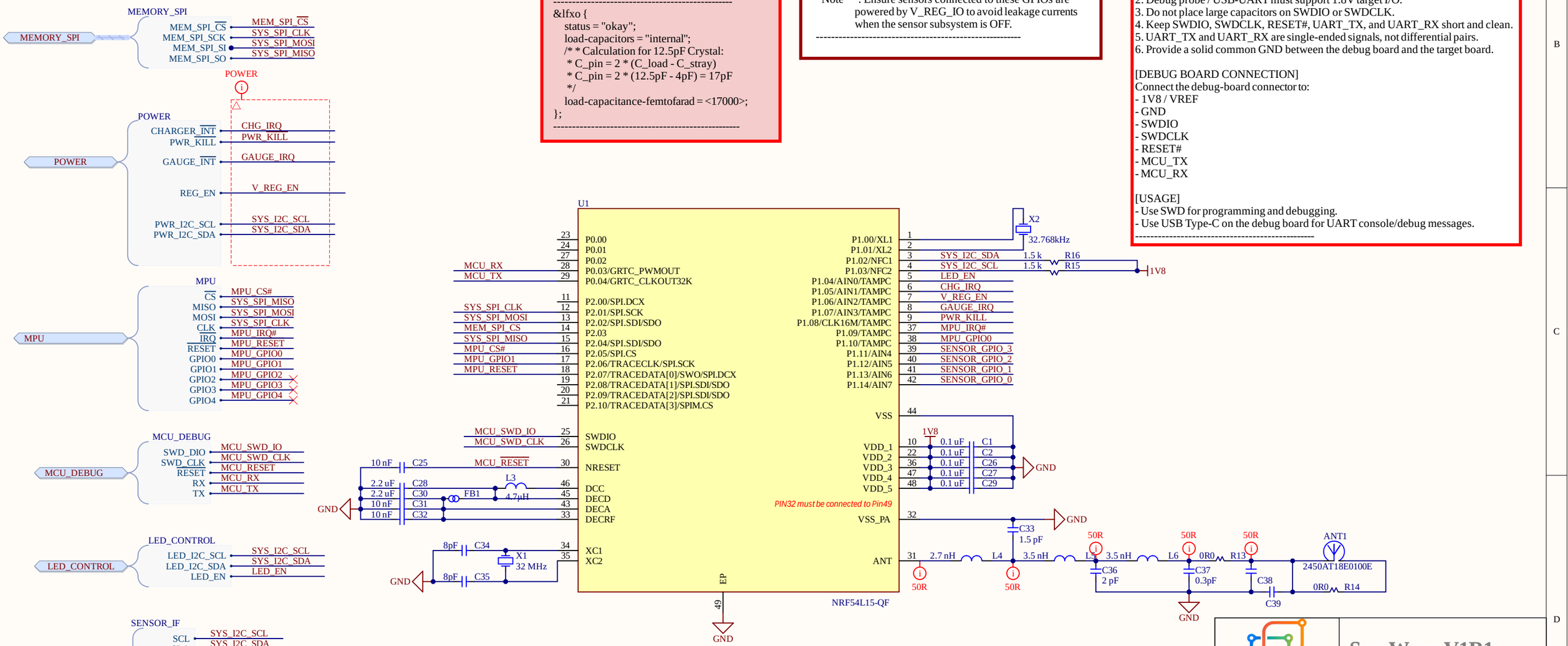
[SWD PROGRAMMING]
SWDIO : SWDIO (Physical Pin 25)
- Bidirectional Data line.
- Has internal pull-up.
SWDCLK : SWDCLK (Physical Pin 26)
- Clock input from debugger.
- Has internal pull-down.
RESET# : nRESET (Physical Pin 30)
- Active Low System Reset.
- Mandatory for device recovery (un-bricking).

[UART CONSOLE] - Instance: UART30
MCU_RX : P0.03 (Pin 28)
- Input to nRF54. Connect to Debugger TX.
MCU_TX : P0.04 (Pin 29)
- Output from nRF54. Connect to Debugger RX.
NOTE:
P0.03 also has GRTC PWM functionality.
P0.04 also has GRTC CLKOUT32K functionality.
Do not enable those GRTC functions while using these pins for UART.

[LAYOUT REQUIREMENTS]
1. Debug connector VREF/VTref must be connected to 1V8.
2. Debug probe / USB-UART must support 1.8V target I/O.
3. Do not place large capacitors on SWDIO or SWDCLK.
4. Keep SWDIO, SWDCLK, RESET#, UART_TX, and UART_RX short and clean.
5. UART_TX and UART_RX are single-ended signals, not differential pairs.
6. Provide a solid common GND between the debug board and the target board.

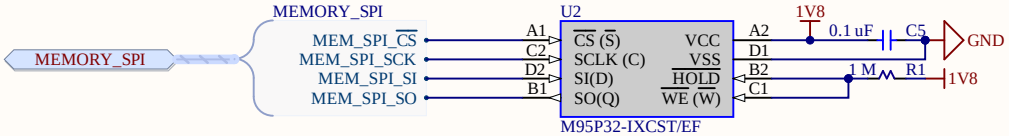
[DEBUG BOARD CONNECTION]
Connect the debug-board connector to:
- 1V8 / VREF
- GND
- SWDIO
- SWDCLK
- RESET#
- MCU_TX
- MCU_RX

[USAGE]
- Use SWD for programming and debugging.
- Use USB Type-C on the debug board for UART console/debug messages.



Designer/Engineer: Yusein Ali	Board Title: SensWear-Hub		
Drawn By: Yusein Ali	Document Name: SensWear_MCU.SchDoc		
Approved By: Dariush Salami	Rev.: C0	Sheet 2 of 8	Size A3
		Date: 31.08.2026	

SensWear-V1R1



EXTERNAL MEMORY (EEPROM) USAGE NOTES
Part: M95P32-IXCST (STMicroelectronics)
Logic Level: 1.8V

[1. INTERFACE CONFIGURATION]
- Protocol : SPI
- Instance : SPIM00 (Shared Bus on Port 2)
- Max Speed: 80 MHz (Device limit) / 32 MHz (MCU limit)
- Mode : CPOL=0, CPHA=0 (Mode 0)
- Bit order: MSB first

[2. PIN MAPPING]

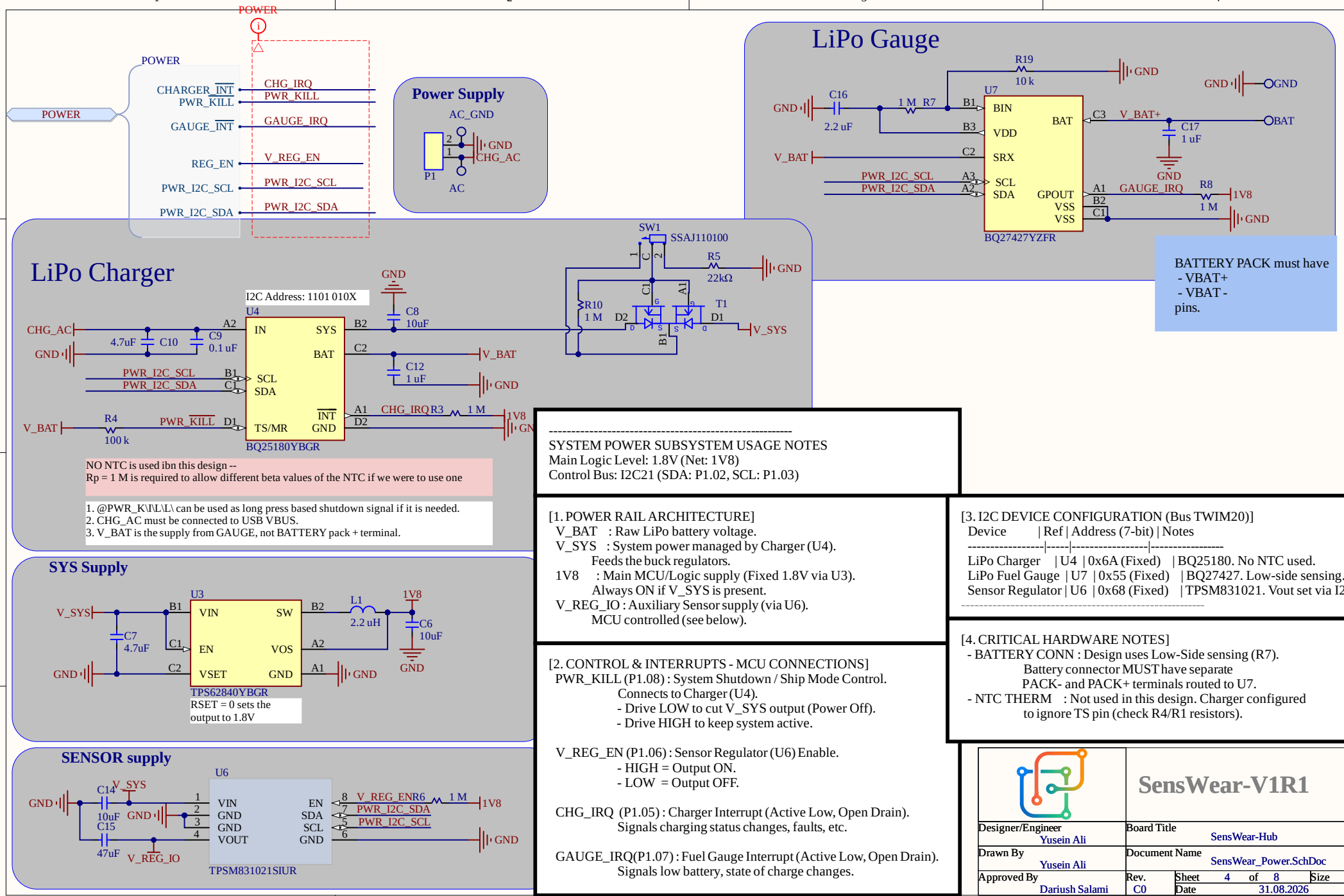
Signal	nRF Pin	Mem Pin	Function
SPI SCK	P2.01	Pin C2	Serial Clock
SPI MOSI	P2.02	Pin D2	Serial Data In (SI)
SPI MISO	P2.04	Pin B1	Serial Data Out (SO)
MEM_CS#	P2.03	Pin A2	Chip Select (Active Low)

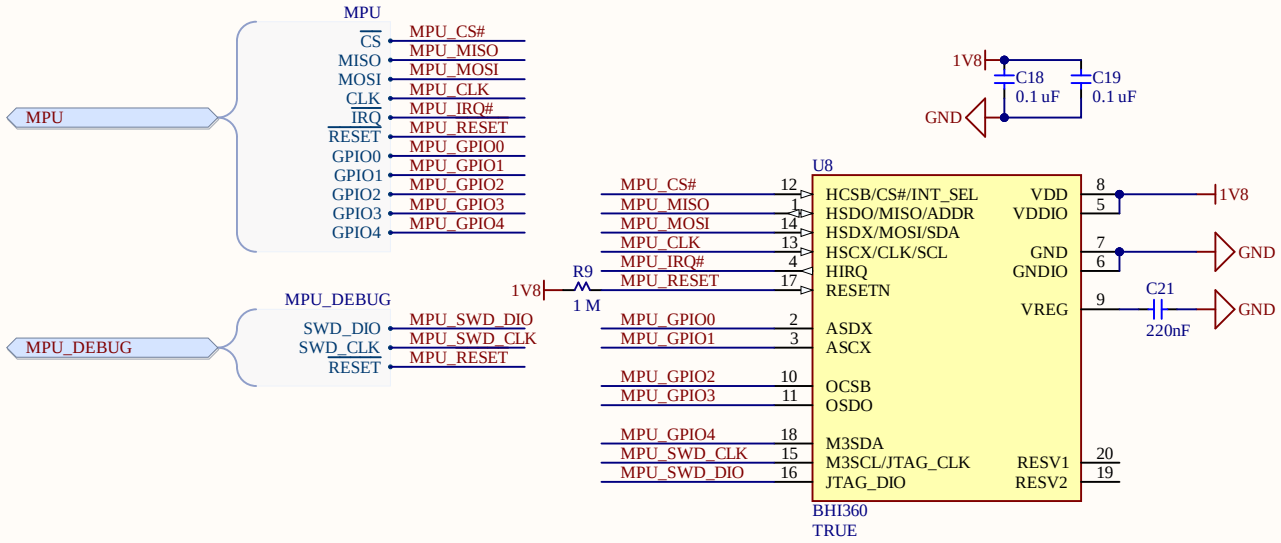
[3. HARDWARE CONFIGURATION]
- HOLD# (Pin B2) : Pulled HIGH (1MΩ).
  - Device suspend function disabled.
- W# (Pin C1) : Pulled HIGH (1MΩ).
  - Hardware Write Protection disabled.
  - Memory is always writable.

[4. POWER & DECOUPLING]
- Supply : 1V8 (Main System Rail).
- Cap : 0.1uF (C5) placed near Pin A2/D1.

[5. PAGE WRITE NOTE]
The M95P32 is a Page EEPROM.
- Page Size: 512 Bytes.
- Unlike standard EEPROMs, it supports very fast Page Write and Buffer Load instructions.

	SensWear-V1R1		
Designer/Engineer: Yusein Ali	Board Title: SensWear-Hub		
Drawn By: Yusein Ali	Document Name: SensWear_Memory.SchDoc		
Approved By: Dariush Salami	Rev.: C0	Sheet: 3 of 8	Size: A4
		Date: 31.08.2026	


POWER
POWER
CHARGER_INT
PWR_KILL
GAUGE_INT
REG_EN
PWR_I2C_SCL
PWR_I2C_SDA
CHG_IRQ
PWR_KILL
GAUGE_IRQ
V_REG_EN
PWR_I2C_SCL
PWR_I2C_SDA
Power Supply
AC_GND
P1
GND
CHG_AC
AC
LiPo Gauge
R19
10 k
GND
C16
2.2 uF
1 M R7
U7
BIN
VDD
SRX
SCL
SDA
BAT
GPOUT
VSS
VSS
BQ27427YZFR
C3
V_BAT+
C17
1 uF
BAT
V_BAT
PWR_I2C_SCL
PWR_I2C_SDA
GAUGE_IRQ
R8
1 M
1V8
BATTERY PACK must have
- VBAT+
- VBAT -
pins.
LiPo Charger
I2C Address: 1101 010X
U4
IN
SYS
BAT
SCL
SDA
INT
TS/MR
GND
BQ25180YBGR
CHG_AC
GND
4.7uF
C10
C9
0.1 uF
C8
10uF
C12
1 uF
V_BAT
R4
100 k
PWR_KILL
CHG_IRQ
R3
1 M
SW1
SSAJ110100
R5
22kΩ
R10
1 M
D2
T1
D1
V_SYS
NO NTC is used ibn this design --
Rp = 1 M is required to allow different beta values of the NTC if we were to use one
1. @PWR_K\I\L\L\ can be used as long press based shutdown signal if it is needed.
2. CHG_AC must be connected to USB VBUS.
3. V_BAT is the supply from GAUGE, not BATTERY pack + terminal.
SYS Supply
U3
VIN
SW
EN
VOS
VSET
GND
TPS62840YBGR
RSET = 0 sets the output to 1.8V
V_SYS
C7
4.7uF
L1
2.2 uH
1V8
C6
10uF
SENSOR supply
U6
VIN
GND
GND
VOUT
EN
SDA
SCL
GND
TPSM831021SIUR
C14
10uF
C15
47uF
V_REG_IO
V_REG_EN
R6
1 M
PWR_I2C_SDA
PWR_I2C_SCL
SYSTEM POWER SUBSYSTEM USAGE NOTES
Main Logic Level: 1.8V (Net: 1V8)
Control Bus: I2C21 (SDA: P1.02, SCL: P1.03)
[1. POWER RAIL ARCHITECTURE]
V_BAT : Raw LiPo battery voltage.
V_SYS : System power managed by Charger (U4).
Feeds the buck regulators.
1V8 : Main MCU/Logic supply (Fixed 1.8V via U3).
Always ON if V_SYS is present.
V_REG_IO : Auxiliary Sensor supply (via U6).
MCU controlled (see below).
[2. CONTROL & INTERRUPTS - MCU CONNECTIONS]
PWR_KILL (P1.08) : System Shutdown / Ship Mode Control.
Connects to Charger (U4).
- Drive LOW to cut V_SYS output (Power Off).
- Drive HIGH to keep system active.
V_REG_EN (P1.06) : Sensor Regulator (U6) Enable.
- HIGH = Output ON.
- LOW = Output OFF.
CHG_IRQ (P1.05) : Charger Interrupt (Active Low, Open Drain).
Signals charging status changes, faults, etc.
GAUGE_IRQ(P1.07) : Fuel Gauge Interrupt (Active Low, Open Drain).
Signals low battery, state of charge changes.
[3. I2C DEVICE CONFIGURATION (Bus TWIM20)]
Device | Ref | Address (7-bit) | Notes
LiPo Charger | U4 | 0x6A (Fixed) | BQ25180. No NTC used.
LiPo Fuel Gauge | U7 | 0x55 (Fixed) | BQ27427. Low-side sensing.
Sensor Regulator | U6 | 0x68 (Fixed) | TPSM831021. Vout set via I2C.
[4. CRITICAL HARDWARE NOTES]
- BATTERY CONN : Design uses Low-Side sensing (R7).
Battery connector MUST have separate
PACK- and PACK+ terminals routed to U7.
- NTC THERM : Not used in this design. Charger configured
to ignore TS pin (check R4/R1 resistors).
SensWear-V1R1
Designer/Engineer
Yusein Ali
Board Title
SensWear-Hub
Drawn By
Yusein Ali
Document Name
SensWear_Power.SchDoc
Approved By
Dariush Salami
Rev.
C0
Sheet
4
of
8
Size
A4
Date
31.08.2026



MOTION SENSOR (BHI360) USAGE NOTES
Logic Level: 1.8V (VDD_1 / 1V8)

[1. INTERFACE CONFIGURATION]
- Protocol : SPI (4-Wire)
- Instance : SPI00 (High-Speed Port 2 Domain)
- Speed : Up to 20 MHz supported (Init @ 2 MHz recommended)
- Mode : CPOL=0, CPHA=0 (Mode 0) or Mode 3

[2. PIN MAPPING (nRF54 <-> BHI360)]

Signal	nRF Pin	BHI Pin	Function
SPI SCK	P2.01	Pin 13	Serial Clock
SPI MOSI	P2.02	Pin 14	Master Out / Slave In
SPI MISO	P2.04	Pin 1	Master In / Slave Out
MPU_CS#	P2.05	Pin 12	Chip Select (Active Low)
MPU_IRQ#	P1.09	Pin 4	Host Interrupt (Active Low)
MPU_RESET	P2.07	Pin 17	Reset (Active Low)

[3. AUXILIARY GPIOs]
- MPU_GPIO0 P1.10 Pin 2 (Programmable)
- MPU_GPIO1 P2.06 Pin 3 (Programmable)

MOTION SENSOR (BHI360) USAGE NOTES
Logic Level: 1.8V (VDD_1 / 1V8)

[4. BOOT & FIRMWARE LOAD SEQUENCE]
1. Power On: 1V8 Rail stable.
2. Reset: Pulse MPU_RESET (P2.07) LOW for >10us, then HIGH.
3. Wait: Delay 2ms for internal BHI boot.
4. Detect: Read 'Host Interface Control' register.
5. Upload: Burst write firmware RAM patch over SPI.
   (Sensor does not function without this RAM patch).

[5. DEBUG NOTE]
JTAG Pins (15/16) are routed to the debug connector.
They are NOT connected to the nRF54 MCU.

BHI360 SENSOR DEBUGGING (JTAG)
Interface: 2-Wire JTAG (Compact JTAG)
Logic Level: 1.8V

[HARDWARE CONNECTION]
Connector: P2 (10-pin Debug Header)

Pin 9 : JTAG_DIO (Bi-directional Data)
- Net: MPU_SWD_DIO
- Connected to BHI360 Pin 16

Pin 10 : JTAG_CLK (Clock)
- Net: MPU_SWD_CLK
- Connected to BHI360 Pin 15

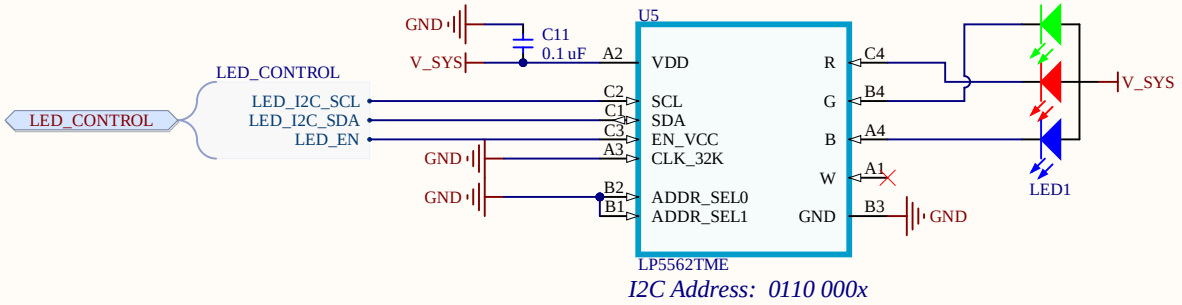
Pin 8 : RESET# (Active Low)
- Net: MPU_RESET
- Connected to BHI360 Pin 17

[CRITICAL SAFETY NOTE]
The MPU_RESET line is shared with the nRF54 MCU (Pin P2.07).

BEFORE connecting a JTAG debugger to the sensor:
1. The nRF54 firmware MUST configure P2.07 as a High-Impedance INPUT.
2. If the nRF54 drives this pin HIGH while the debugger pulls it LOW, you risk a short circuit damaging the MCU or Sensor.

[DEBUGGER SETTINGS]
Target Device : Bosch BHI360 (ARC EM4 Core)
Protocol : cJTAG (Compact JTAG / 2-wire)
Speed : 1 MHz - 4 MHz (Recommend starting slow)

	SensWear-V1R1	
Designer/Engineer: Yusein Ali	Board Title: SensWear-Hub	
Drawn By: Yusein Ali	Document Name: SensWear_MPU.SchDoc	
Approved By: Dariush Salami	Rev.: C0	Sheet 5 of 8 / Size A4 / Date 31.08.2026



RGBW LED DRIVER (LP5562)
Logic Level: 1.8V

[CONTROL INTERFACE]
I2C Addr : 0x30 (7-bit)
(Requires A1=GND, A0=GND)
Bus : I2C21 (SDA: P1.02, SCL: P1.03)

[HARDWARE ENABLE]
LED_EN : P1.04 (Pin 17 on nRF54)
Polarity : Active HIGH.

CRITICAL FIRMWARE NOTE:
1. Set P1.04 HIGH.
2. Wait 500µs (T_startup).
3. ONLY THEN attempt I2C communication.
(Chip ignores I2C when EN is Low).

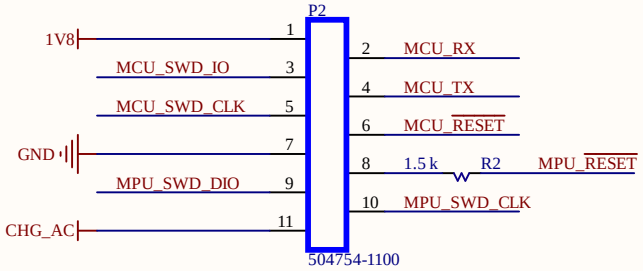
[OUTPUT CHANNELS]
Type : Current Sinks (Connect LED Cathodes to pins).
R, G, B : Programmable (PWM or Pattern Engine).
W (White): Direct PWM control (often used for status).

[POWER SAVE]
To enter absolute lowest power (Shutdown):
Drive LED_EN (P1.04) -> LOW.

	SensWear-V1R1			
Designer/Engineer: Yusein Ali	Board Title: SensWear-Hub			
Drawn By: Yusein Ali	Document Name: SensWear_RGB.SchDoc			
Approved By: Dariush Salami	Rev.: C0	Sheet 6 of 8	Size: A4	
		Date: 31.08.2026		







RESET LINE PROTECTION (MPU_RESET)

Component : Resistor (Series)
Value : 1.5kΩ
Location : Series between nRF54 Pin P2.07 and the Debug Connector (Pin 8).
Function : - Prevents short-circuit damage if nRF54 accidentally drives line HIGH while Debugger drives LOW.
- Current limited to ~1.2mA.

MPU DEBUG INTERFACE (BHI360)
Logic Level: 1.8V (VDDIO)

[JTAG / SWD CONNECTION] - Connector P2
MPU_SWD_DIO : Pin 9
- Connected to BHI360 Pin 16 (JTAG_DIO)
- Bi-directional Data
MPU_SWD_CLK : Pin 10
- Connected to BHI360 Pin 15 (JTAG_CLK)
- Clock Input
MPU_RESET# : Pin 8
- Active Low Reset
- Connected to BHI360 Pin 17 (RESETN)

[CRITICAL RESET NOTE]
The line 'MPU_RESET' is shared between:
1. External Debug Connector (Pin 8)
2. nRF54 MCU Pin P2.07 (Pin 7 on U1)

WARNING: To debug the BHI360 via Connector P2, the nRF54 MCU must configure P2.07 as INPUT (High-Z) to avoid driving the line HIGH while the debugger tries to pull it LOW.

[DEBUGGER COMPATIBILITY]
Target Core : ARC EM4 (Fuser Core)
Interface : 2-Wire JTAG/SWD

MCU (nRF54L15)DEBUG & DIAGNOSTIC INTERFACE
Logic Level: 1.8V (VDD_IO)

[SWD PROGRAMMING]
SWDIO : SWDIO (Physical Pin 25)
- Bidirectional Data line.
- Has internal pull-up.
SWDCLK : SWDCLK (Physical Pin 26)
- Clock input from debugger.
- Has internal pull-down.
RESET# : nRESET (Physical Pin 30)
- Active Low System Reset.
- Mandatory for device recovery (un-bricking).

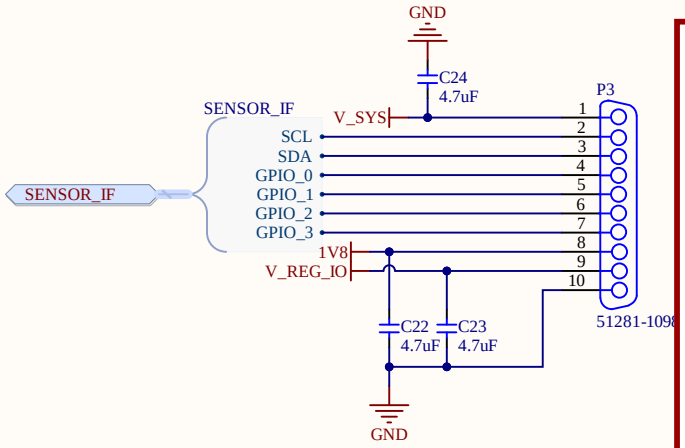
[UART CONSOLE] - Instance: UART30
MCU_RX : P0.03 (Pin 28)
- Input to nRF54. Connect to Debugger TX.
MCU_TX : P0.04 (Pin 29)
- Output from nRF54. Connect to Debugger RX.
NOTE:
P0.03 also has GRTC PWM functionality.
P0.04 also has GRTC CLKOUT32K functionality.
Do not enable those GRTC functions while using these pins for UART.

[LAYOUT REQUIREMENTS]
1. Debug connector VREF/VTref must be connected to 1V8.
2. Debug probe / USB-UART must support 1.8V target I/O.
3. Do not place large capacitors on SWDIO or SWDCLK.
4. Keep SWDIO, SWDCLK, RESET#, UART_TX, and UART_RX short and clean.
5. UART_TX and UART_RX are single-ended signals, not differential pairs.
6. Provide a solid common GND between the debug board and the target board.

[DEBUG BOARD CONNECTION]
Connect the debug-board connector to:
- 1V8 / VREF
- GND
- SWDIO
- SWDCLK
- RESET#
- MCU_TX
- MCU_RX

[USAGE]
- Use SWD for programming and debugging.
- Use USB Type-C on the debug board for UART console/debug messages.

	SensWear-V1R1		
Designer/Engineer: Yusein Ali	Board Title: SensWear-Hub		
Drawn By: Yusein Ali	Document Name: SensWear_Debug.SchDoc		
Approved By: Dariush Salami	Rev.: C0	Sheet: 7 of 8	Size: A4
		Date: 31.08.2026	



SENSOR POWER REGULATOR (U6)
Part: TPSM831021SIUR

[HARDWARE CONFIGURATION]
Input : V_SYS (System Voltage)
Output : V_REG_IO (Sensor Logic Supply)
I2C Addr : 0x68 (7-bit)
(Factory programmed, VOUT configurable)

[CONTROL LOGIC]
ENABLE : V_REG_EN (P1.06)
- Active HIGH.
- Drive HIGH to power on sensors.
- Drive LOW to save power (0.1uA shutdown).

[FIRMWARE SEQUENCE]
1. Set P1.06 HIGH.
2. Wait 2ms (Soft-start time).
3. Initialize I2C sensors on the V_REG_IO rail.

[LAYOUT NOTE]
- 10uF Input Cap (C14) must be close to VIN.
- 47uF Output Cap (C15) must be close to VOUT.

SENSOR INTERFACE (GENERIC)
Logic Level: 1.8V (V_REG_IO Domain)

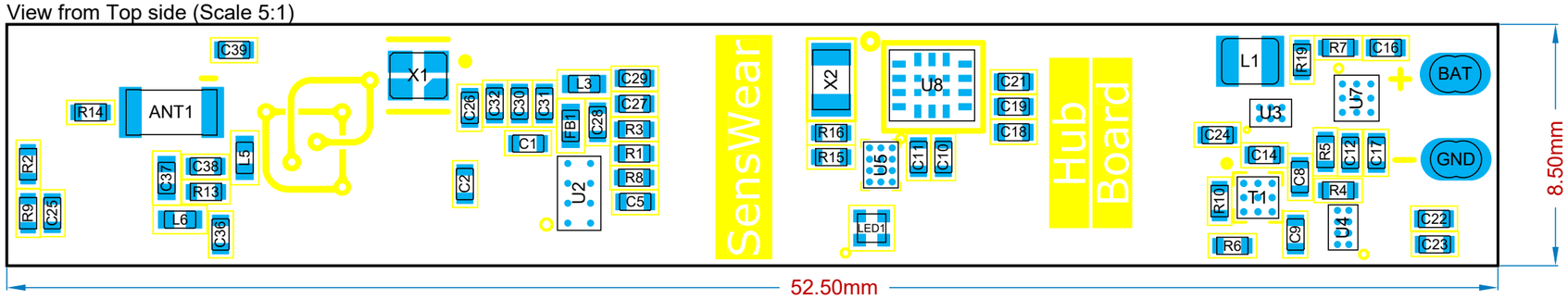
[SHARED CONTROL BUS]
Interface : I2C (I2C21)
SDA : P1.02 (Pin 3)
SCL : P1.03 (Pin 4)
Pull-ups : 1.5kΩ to 1V8 (R15, R16) installed on board.

[SENSOR GPIOs / ANALOG INPUTS]
GPIO_0 : P1.14 / AIN7 (Pin 42)
- Capable of Analog Input (SAADC).
GPIO_1 : P1.13 / AIN0 (Pin 41)
- Capable of Analog Input (SAADC).
GPIO_2 : P1.12 / AIN1 (Pin 40)
- Capable of Analog Input (SAADC).
GPIO_3 : P2.11 (Pin 39)
- Capable of Analog Input (SAADC).

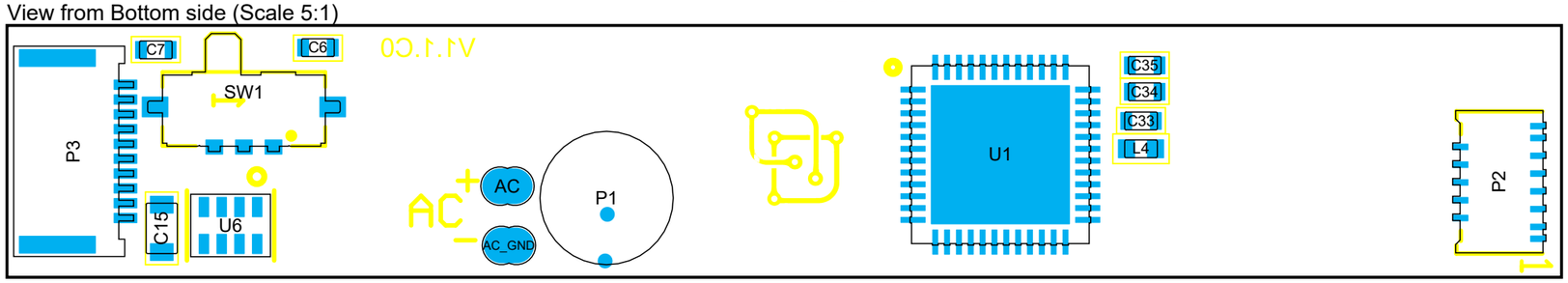
[POWER DOMAIN]
Supply : V_REG_IO (Switched Rail)
Control : V_REG_EN (P1.06) must be HIGH.
Note : Ensure sensors connected to these GPIOs are powered by V_REG_IO to avoid leakage currents when the sensor subsystem is OFF.

	SensWear-V1R1			
Designer/Engineer: Yusein Ali	Board Title: SensWear-Hub			
Drawn By: Yusein Ali	Document Name: SensWear_DaughterBoard.SchDoc			
Approved By: Dariush Salami	Rev.: C0	Sheet 8 of 8	Size: A4	
		Date: 31.08.2026		

View from Top side (Scale 5:1)



View from Bottom side (Scale 5:1)



Bill Of Materials

Line #	Designator	Quantity	Manufacturer	Manufacturer Part Number
1	ANT1	1	Johanson	2450AT18E0100E
2	C1, C2, C5, C9, C11, C18, C19, C26, C27, C29	10	Murata Electronics	GRM155R61C104KA88D
3	C6, C8, C14	3	Samsung	CL05A106MP5NUNC
4	C7, C10, C22, C23, C24	5	Murata	GRM155R61A475MEAAD
5	C12, C17	2	Murata Electronics North America	GRM155R61A105KE15D
6	C15	1	Samsung	CL10A476MQ8QRNC
7	C16, C28, C30	3	Murata Electronics North America	GRM155R61A225KE95D
8	C21	1	Murata	GRM155R61C224KA12D
9	C25, C31, C32	3	Murata Electronics North America	GRM155R71H103KA88D
10	C33	1	Murata	GJM1555C1H1R5BB01D
11	C34, C35	2	Murata	GRM1555C1H8R0BA01D
12	C36	1	Murata	GJM1555C1H2R0WB01D
13	C37	1	Murata	GJM1555C1HR30WB01D
14	C38, C39	2		
15	FB1	1	Murata	BLM15AG121SN1D
16	L1	1	Murata	DFE201612E-2R2M=P2
17	L3	1	TDK	PLEA67BCA4R7M-1PT00

Bill Of Materials

Line #	Designator	Quantity	Manufacturer	Manufacturer Part Number
18	L4	1	Murata	LQG15WH2N7B02D
19	L5, L6	2	Murata	LQG15WH3N5B02D
20	LED1	1	Lumex	SML-LX0404SIUPGUSB
21	P1	1	XTPogoPin	RM-1238-B02-41
22	P2	1	Molex	504754-1100
23	P3	1	Molex	51281-1098
24	R1, R3, R6, R7, R8, R9, R10	7	Panasonic Electronic Components	ERJ-2RKF1004X
25	R2, R15, R16	3	Panasonic Electronic Components	ERJ-2RKF1501X
26	R4	1	Panasonic Electronic Components	ERJ-2RKF1003X
27	R5	1	Panasonic	ERJ2RKF2202X
28	R13, R14	2	Panasonic Electronic Components	ERJ-2GE0R00X
29	R19	1	Panasonic Electronic Components	ERJ-2RKF1002X
30	SW1	1	ALPS	SSAJ110100
31	T1	1	Texas Instruments	CSD75207W15
32	U1	1	Nordic Semiconductor	NRF54L15-QFAA-R
33	U2	1	STMicroelectronics	M95P32-IXCST/EF
34	U3	1	Texas Instruments	TPS62840YBGR
35	U4	1	Texas Instruments	BQ25180YBGR
36	U5	1	Texas Instruments	LP5562TME/NOPB
37	U6	1	Texas Instruments	TPSM831021SIUR
38	U7	1	Texas Instruments	BQ27427YZFR
39	U8	1	Bosch Tools	BHI360

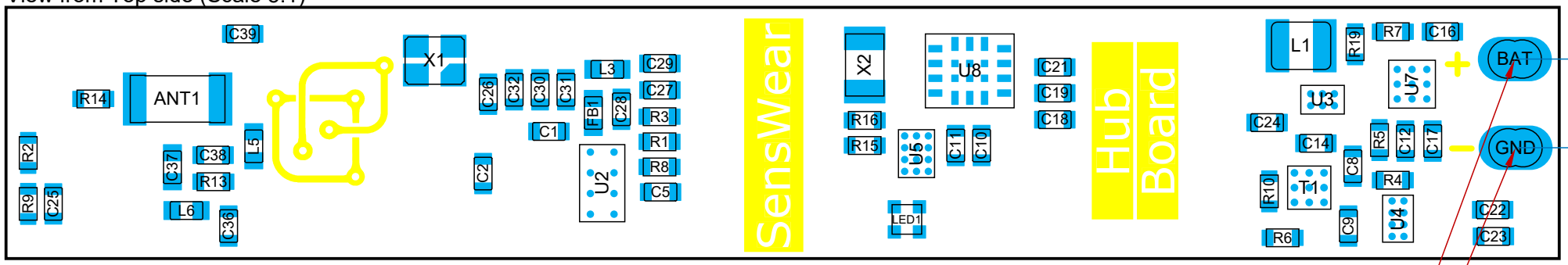
Assembly steps:

1. The boards are assembled without the battery including the POGO connector P1: XTPogoPin, RM-1238-B02-41
2. Battery is not assembled
3. P1 is magnetized
4. The battery is assembled.

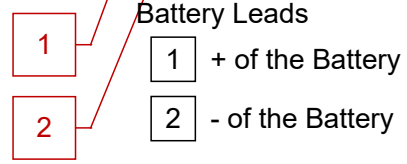
SensWear-V1R1

Designer/Engineer	Board Title		
Yusein Ali	SensWear-Hub		
Drawn By	**Document Name**		
Yusein Ali	SensWear-Hub-assembly.PCBDwf		
Approved By	**Ver.**	**Rev.**	**Date**
Dariush Salami	1	C0	31.08.2026

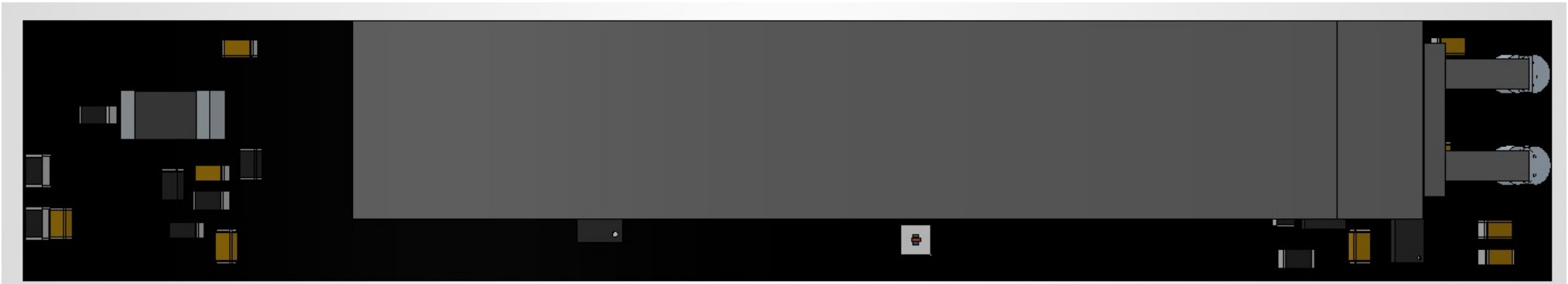
View from Top side (Scale 5:1)

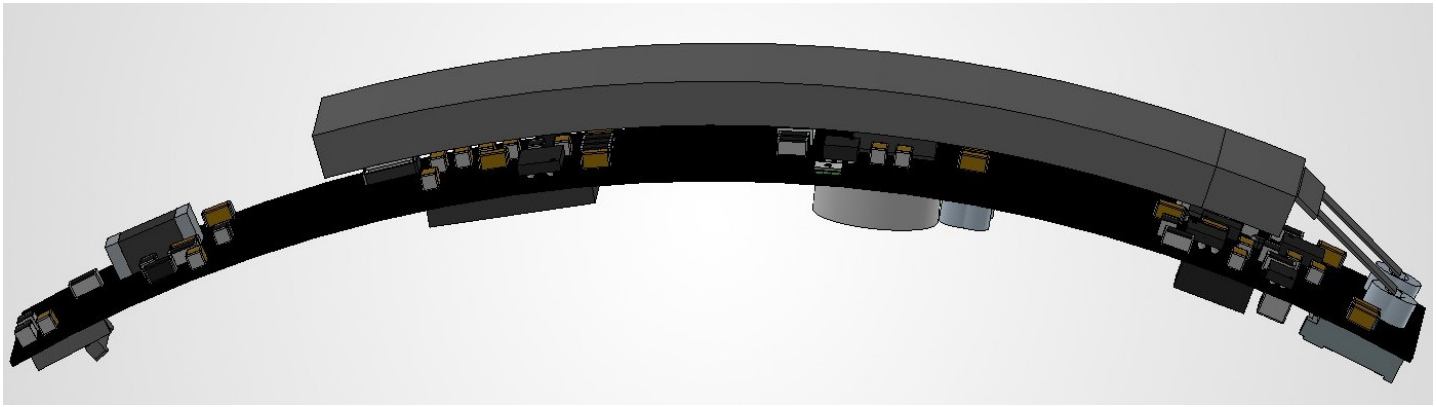


Curved batteries can be welded by soldering or laser welding.
- Soldering is done at 360 degrees and completed within 3 seconds.
- Laser welding needs to be tuned according to the assembly factory's welding equipment, with no universal standard. Most importantly, the welding temperature and duration must be well controlled.



The POGO connector P1 (XTPogoPin, RM-1238-B02-41) MUST be magnetized before battery is assembled!





SensWear-V1R1			
Designer/Engineer Yusein Ali	**Board Title** SensWear-Hub		
Drawn By Yusein Ali	**Document Name** SensWear-V1R1-MainBoard-		
Approved By Dariush Salami	**Ver.** 1	**Rev.** C0	**Date** 31.08.2026